

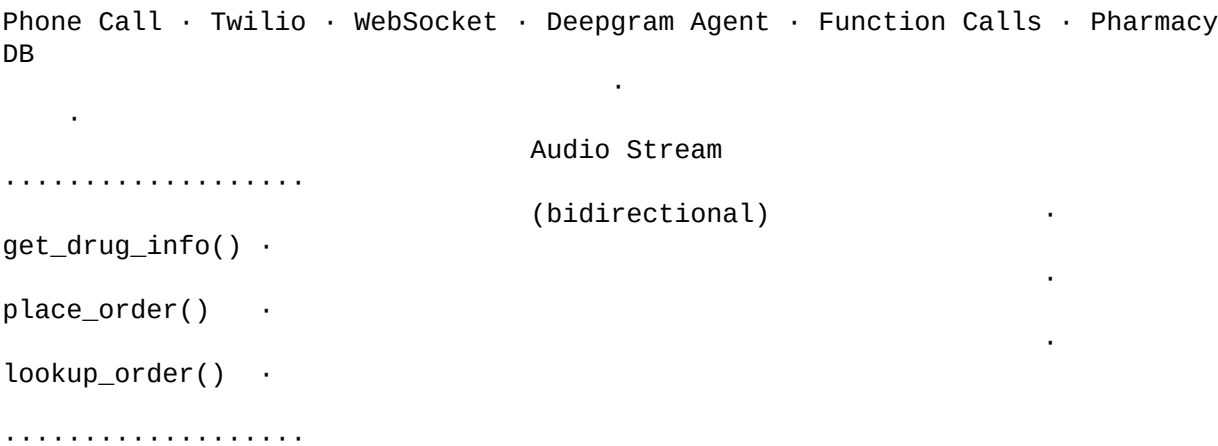
Deepgram Voice Agent

A **real-time voice-based pharmacy assistant** using **Deepgram's agentic AI** with **Twilio** integration for phone call handling.

· Features

- **Real-time voice interaction** via WebSocket audio streaming
 - **Pharmacy operations** · Drug info, order placement, order lookup
 - **Twilio integration** · Handle actual phone calls
 - **Barge-in support** · User can interrupt the agent mid-response
 - **Function calling** · Agent triggers domain-specific functions
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.. Architecture



· File Structure

| File | Purpose | |-----|-----| | `main.py` | WebSocket handlers for Twilio · Deepgram integration (177 lines) |
| `pharmacy_functions.py` | Pharmacy function handlers: drug info, orders (88 lines) | | `config.json` | Agent configuration and settings | | `pyproject.toml` | Dependencies |

· Getting Started

Prerequisites

- Python 3.10+, Deepgram API key, Twilio account

```
cd DeepgramVoiceAgent
uv sync
```

Environment Variables (.env)

```
DEEPGRAM_API_KEY=...
TWILIO_ACCOUNT_SID=...
TWILIO_AUTH_TOKEN=...
```

Run

```
uv run python main.py
```

Starts WebSocket server on `localhost:5000`.

· Logic Flow

- 1 **Phone call** arrives via Twilio
- 2 **WebSocket** connection established between Twilio and server
- 3 **Audio stream** forwarded to Deepgram for speech-to-text
- 4 **Agent processes** the transcribed text
- 5 **Function calls** triggered when agent needs pharmacy data
- 6 **Response audio** streamed back via Deepgram TTS · Twilio · caller

Pharmacy Functions

Function	Description
<code>get_drug_info(drug_name)</code>	Look up drug details from in-memory database
<code>place_order(drug, quantity)</code>	Create a new pharmacy order
<code>lookup_order(order_id)</code>	Check status of existing order

· Dependencies

`websockets, deepgram-sdk, twilio, python-dotenv`

· License

Educational project · use freely for learning and reference.